```
# ==========================================
# PCA DIMENSIONALITY REDUCTION
# ==========================================

# Step 1: Import Libraries
from sklearn import datasets
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# ------------------------------------------
# Step 2: Load Iris Dataset
# ------------------------------------------

iris = datasets.load_iris()

df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)

print(df.head())

# ------------------------------------------
# Step 3: Standardize the Data
# ------------------------------------------

scaler = StandardScaler()

scaled_data = scaler.fit_transform(df)

scaled_data = pd.DataFrame(scaled_data)

print("\nScaled Data")
print(scaled_data.head())

# ------------------------------------------
# Step 4: Correlation Heatmap
# ------------------------------------------

plt.figure(figsize=(8,6))

sns.heatmap(
    scaled_data.corr(),
    annot=True
)

plt.title("Correlation Heatmap of Scaled Data")

plt.show()

# ------------------------------------------
# Step 5: Apply PCA
# ------------------------------------------

pca = PCA(n_components=3)

pca.fit(scaled_data)

data_pca = pca.transform(scaled_data)

# Convert into DataFrame
data_pca = pd.DataFrame(
    data_pca,
    columns=['PC1', 'PC2', 'PC3']
)

print("\nPCA Data")
print(data_pca.head())

# ------------------------------------------
# Step 6: Explained Variance Ratio
# ------------------------------------------

print("\nExplained Variance Ratio")
print(pca.explained_variance_ratio_)
```

```
# -----------------------------------------
# Step 7: Scree Plot
# -----------------------------------------

plt.figure(figsize=(7,5))

plt.plot(
    range(1,4),
    pca.explained_variance_ratio_,
    marker='o'
)

plt.xlabel("Principal Components")
plt.ylabel("Explained Variance Ratio")
plt.title("Scree Plot")

plt.show()

# -----------------------------------------
# Step 8: Heatmap of PCA Components
# -----------------------------------------

plt.figure(figsize=(6,5))

sns.heatmap(
    data_pca.corr(),
    annot=True
)

plt.title("Correlation Heatmap of PCA Data")

plt.show()

# -----------------------------------------
# Step 9: Scatter Plot of Reduced Data
# -----------------------------------------

plt.figure(figsize=(8,6))

plt.scatter(
    data_pca['PC1'],
    data_pca['PC2']
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")

plt.title("PCA Reduced Data Visualization")

plt.show()

print()
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```
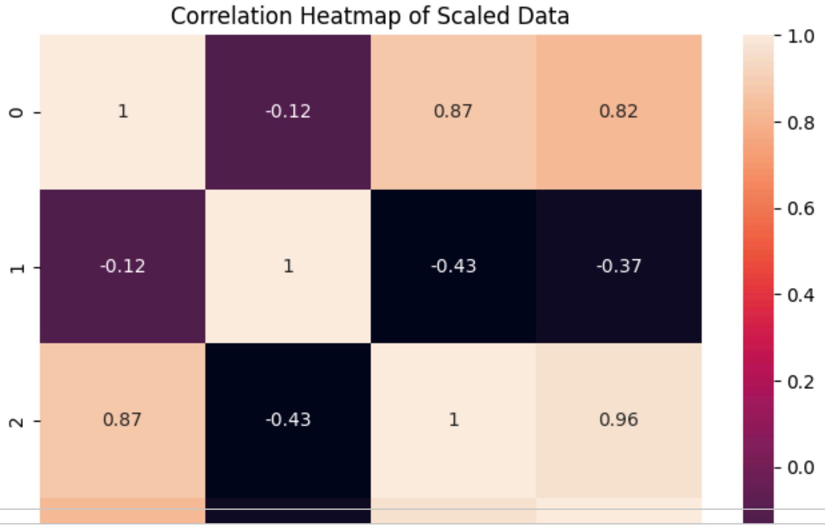

```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

Scaled Data
          0         1         2         3
0 -0.900681  1.019004 -1.340227 -1.315444
1 -1.143017 -0.131979 -1.340227 -1.315444
2 -1.385353  0.328414 -1.397064 -1.315444
3 -1.506521  0.098217 -1.283389 -1.315444
4 -1.021849  1.249201 -1.340227 -1.315444
```




```
print()
print()
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print()
print()
```

```
PCA Data
        PC1       PC2       PC3
0 -2.264703  0.480027  0.127706
1 -2.080961 -0.674134  0.234609
2 -2.364229 -0.341908 -0.044201
3 -2.299384 -0.597395 -0.091290
4 -2.389842  0.646835 -0.015738
```

```
print("Completed")
```

```
Completed
```